

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total) Add your name to the file name (\_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

**Abstract of Scientific Justification (300 words max) (2 points):** Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

We will take spectroscopic data on a supernova with the “B&C” Spectrograph on the Bok 90” telescope at Kitt Peak. Due to the transient, quickly fading nature of supernovae, it is optimal to observe the target within 1-2 months of peak magnitude. Therefore it is uncertain surrounding the exact scientific goal of the project until the observations are taken on October 17-19. However, we can anticipate that spectra may allow us to categorize the supernova if it is caught early enough, which would be a significant scientific result. The method for analysing data will be the SNID-Sage software, which enables us to identify diagnosable features of supernova types, such as the presence and strength of Hydrogen and/or Helium lines (Type I and II and subtypes). With the SNID-Sage software, we can also fit our data and calculate the radial velocity of the target from the Doppler shift of spectral lines, indicating binarity/multiplicity. We give an outline and possible targets, but the actual target and analysable data will be determined at the time of observation.

**Telescope Requested (0.5 points): BOK 90”**

**Instrument (0.5 points): “B&C” Spectrograph**

**Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points):** Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

At the time of our observation (Oct. 17-19), the sun will have an RA of around 14 hours, while the moon's RA will be between 18 and 20. This means the moon will be reaching its peak right around sunset, however we have estimated the brightness on these dates with Catalina Sky Survey, which will be ranging from 38% to 58% illuminated as the weekend progresses, with the minimum on our first possible observing day, the 17th. It would therefore be ideal to observe on the 17th, when we will have the least interference from the moon with our observations. The moon will set between 11:00PM and 1:00AM on all of these dates, so a target with an RA around 22-3 hours would be also ideal as these will be crossing the meridian while both the sun and moon are below the horizon and we will therefore receive no interference.

### Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Sources used:

[Recently Discovered SN](#)

[Current Magnitude Tracking via Rochester Astro](#)

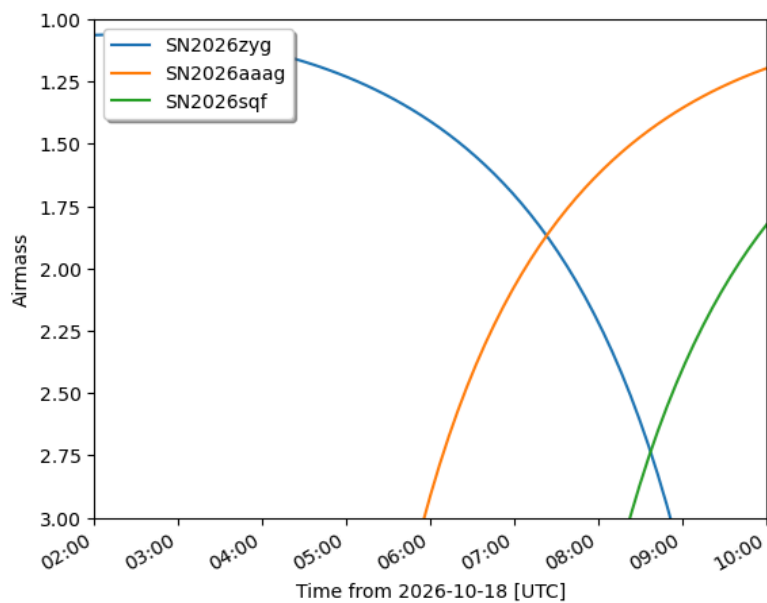
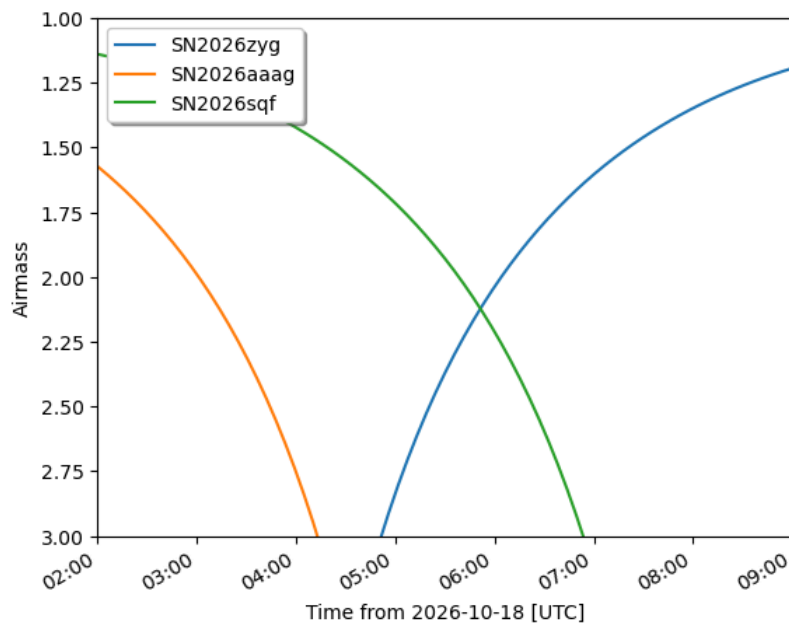
[UTC to Tucson Time](#)

Constellations highly visible in Oct: Cygnus, Pegasus, Cassiopeia, Cephus, Andromeda, Ursa Major/Minor

Constellations only visible at beginning/end of night: Aquarius, Capricornus, Cetus, Cygnus, Aquila, Lyra, Taurus, Auriga, Orion, Lynx

| Object Name | Right Ascension | Declination | V magnitude at peak | Peak Date | Object Type | Visibility |
|-------------|-----------------|-------------|---------------------|-----------|-------------|------------|
| SN 2026sqf  | 140.673888      | 53.37       | 12.9                | 7/28      | SN Ia       | ursa major |
| SN          | 102.05          | 51.23555556 | 18.08               | 8/29      | SN type     | Lynx       |

|                            |                 |                |               |             |              |               |
|----------------------------|-----------------|----------------|---------------|-------------|--------------|---------------|
| <b>2026aaag</b>            |                 |                |               |             | <b>SN II</b> |               |
| <a href="#">SN 2026zyg</a> | <b>308.8761</b> | <b>51.5900</b> | <b>18.049</b> | <b>8/28</b> | <b>SN II</b> | <b>Cygnus</b> |

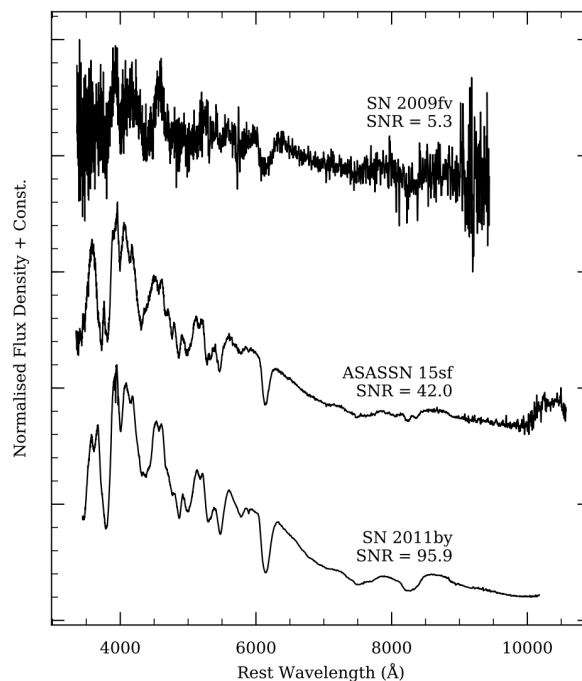


Caption: Airmass plot for the targets listed above in the table for October 17th. SN2026zyg would be the most ideal of these targets because it reaches peak when the moon is set, and it gets very high and therefore has less airmass, the typical aimed-for value being 1.5 times zenith airmass.

## Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

We will want a minimum signal to noise ratio of five, ideally higher, as while supernovae are indeed very bright objects, we will need the best possible SNR to get spectra we can work with and identify real lines. We show a plot from Stahl et al. (2020) which observed Type Ia supernovae at varying signal to noise ratios with a 3 meter telescope. In the paper they report an SNR of 8.3 for an object observed with an exposure time of 1200 seconds. With a 2.3 meter telescope and likely similarly bright source, we would likely need between 1200 and 2000 seconds of exposure time to get a SNR of 5.



Stahl et al. (2020)

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

Spectra measures isotope absorption and emission lines, which are used in the

categorization of supernovae based on the abundances and presence of elements like Hydrogen and Helium. In the case that there are no Hydrogen absorption features in the spectrum, it is Type I, and furthermore the ionized Silicon absorption feature is an indicator of a Type Ia supernova specifically. Fitting spectra with SNID-Sage allows us to easily identify these features, while calculations of optical depth can give us an idea of the abundances of certain elements. We can also estimate the radial velocity from the Doppler shift of certain elemental features with a known rest wavelength.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If we are unable to gather our own spectroscopic data for a multitude of possible setbacks, we may be able to use similar observations from projects that have been conducted at Steward Observatory. For example, Professor Nathan Smith does regular supernova spectroscopy using the same instrument, so we could ask him to share some of his recent observations and classify/analyse them.